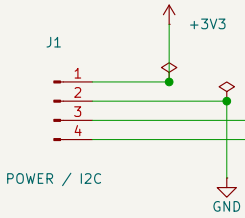
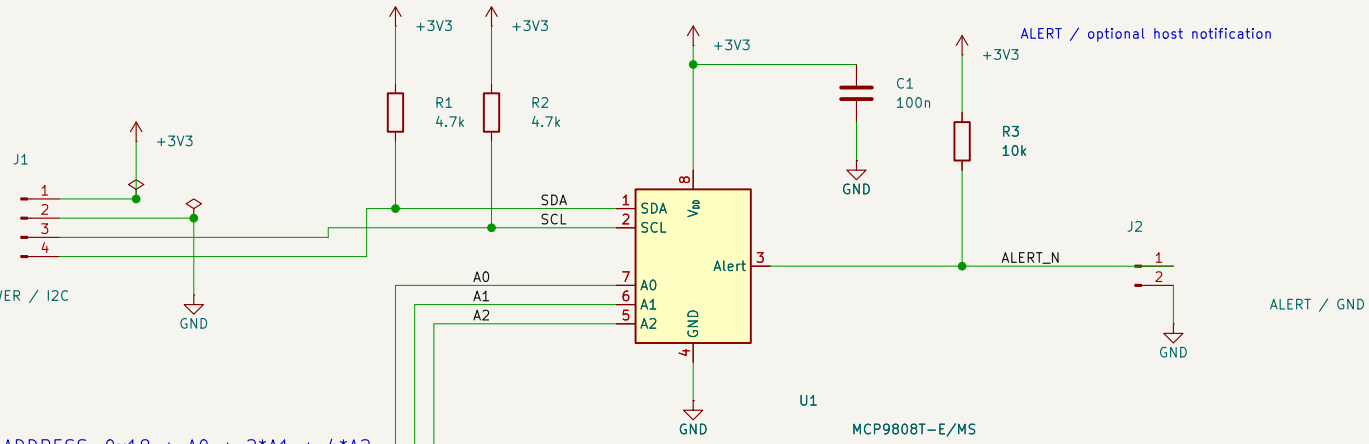
01 / HOST + BUS

3.3 V +/−5% only; common−ground host

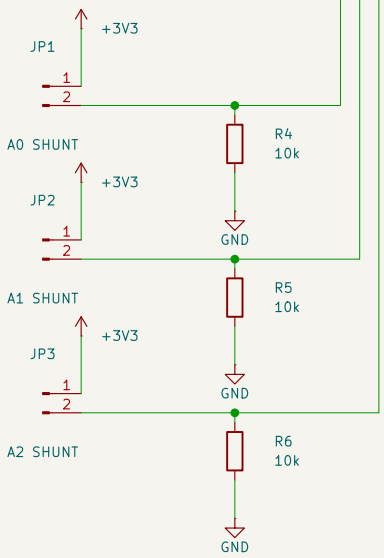


02 / SENSOR + LOCAL BYPASS



03 / ADDRESS: 0x18 + A0 + 2*A1 + 4*A2

No shunts fitted = 0x18. Power off before changing shunts.



OPERATING ASSUMPTIONS

100 kHz initial bus; scope rise time with real cable / pullups.
Fixed 4.7k pullups: assess parallel host pullups before use.
ALERT is open drain; disabled until firmware enables it.
100n local bypass: bulk capacitance supplied by host.
No ESD / level shifting / reverse−polarity protection.
Unbuilt DRAFT: temperature accuracy and fit unqualified.
Sources + calculations: DESIGN_BASIS.md, evidence/index.yaml

04 / MECHANICAL

4 x 3.2 mm NPTH; 45 x 35 mm board



MCP9808T−E/MS; source and assumptions in DESIGN_BASIS.md
Unbuilt / not approved for fabrication
PCB Copilot examples

Sheet: /
File: i2c−sensor.kicad_sch

Title: I2C temperature sensor / 3.3 V

Size: A4	Date: 2026−10−29	**Rev: A1 DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1